TERRAFORM FINAL VIEW ANSWERS

1. What is Terraform?
Definition: Terraform is an Infrastructure as Code (IaC) tool used to define, provision, and manage infrastructure using configuration files/code instead of manually creating and managing infrastructure.
Simple meaning:
Terraform means managing infrastructure through code instead of manually creating resources through a cloud provider's console. Instead of manually creating servers, networks, databases, load balancers, and other infrastructure components, we can define what infrastructure we need in Terraform configuration files.
Terraform reads these configuration files and communicates with the required cloud provider through APIs to create, update, and manage the infrastructure.
Why do we use Terraform?
The main reason we use Terraform is to automate infrastructure creation and management.
If infrastructure is created manually, the same steps may need to be repeated for different environments or applications. This can take more time and can lead to configuration mistakes.
Terraform allows us to define infrastructure as code and reuse the same configuration whenever required.
This gives us:

- **Automation** — Reduces manual infrastructure creation.
- **Repeatability** — The same infrastructure can be created again.
- **Consistency** — Different environments can follow the same infrastructure design.
- **Version Control** — Terraform code can be stored and managed in Git.
- **Easy Changes** — Infrastructure changes can be made by modifying the configuration.
- **Review** — Infrastructure changes can be reviewed before they are applied.
- **Multi-cloud support** — Terraform can manage platforms such as AWS, Azure, Google Cloud, Kubernetes, and many other services.

When do we use Terraform?
Terraform is especially useful when infrastructure is:

- Large
- Complex
- Repeated frequently
- Managed by a team
- Required across multiple environments
- Required to be version-controlled

For a very small temporary test, Terraform may not always be necessary. But when infrastructure becomes larger, repeatable, or team-managed, Terraform becomes much more useful.
How does Terraform work?
Terraform follows the concept of **desired state**.
We define how we want our infrastructure to look in Terraform configuration.
Terraform then evaluates the configuration and the current infrastructure information available to it, determines what changes are required, and applies those changes through the appropriate provider.
The basic flow is:
Terraform Configuration → Terraform → Provider → Cloud Platform → Infrastructure

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2. What is Infrastructure as Code (IaC)?
Definition: Infrastructure as Code (IaC) is the practice of defining, creating, configuring, and managing infrastructure using code or configuration files instead of manually managing infrastructure through a graphical user interface.
Simple meaning:
Infrastructure as Code means treating infrastructure like software code.
Instead of manually creating infrastructure through a cloud console, we write the infrastructure requirements in configuration files. For example, instead of manually creating a network, servers, databases, and load balancers, we define these requirements in code.
The IaC tool reads the configuration and creates the required infrastructure.
Why do we use Infrastructure as Code?
The main purpose of IaC is to make infrastructure creation and management more automated, consistent, repeatable, and easier to maintain.
Without IaC, infrastructure is often created manually. As the environment becomes larger, remembering and repeating every configuration becomes difficult.
With IaC, the infrastructure configuration is stored as code and can be managed like application code.
This provides:

- **Automation** — Infrastructure can be created without repeated manual steps.
- **Consistency** — The same configuration can be used across environments.
- **Repeatability** — Infrastructure can be recreated when required.
- **Version Control** — Infrastructure changes can be tracked in Git.
- **Collaboration** — Multiple team members can work with the same infrastructure code.
- **Reviewability** — Infrastructure changes can be reviewed before deployment.
- **Faster Provisioning** — Large infrastructure can be created more efficiently.

Where is Infrastructure as Code used?
IaC is commonly used in cloud and DevOps environments where infrastructure needs to be created and managed repeatedly.
It can be used to manage:

- Virtual machines
- Networks
- Subnets
- Databases
- Load balancers
- Storage
- Security resources
- Kubernetes infrastructure
- Other cloud resources

Terraform is one of the popular tools used to implement Infrastructure as Code.
How does IaC work?
The required infrastructure is first defined in configuration files.
The IaC tool reads that configuration and communicates with the target platform to create or modify the infrastructure according to the defined configuration.
The basic flow is:
Infrastructure Code → IaC Tool → Cloud Provider → Infrastructure

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3. Why do we use Terraform?
Definition: Terraform is used to automate the creation, modification, and management of infrastructure by defining infrastructure as code.
Simple meaning:
We use Terraform so that we do not have to manually create and configure infrastructure again and again.
Instead, we define the required infrastructure in Terraform configuration files and allow Terraform to manage it.
This is especially useful when an organization has multiple resources and multiple environments such as Development, Testing, and Production.
Why is Terraform useful?
Terraform provides several important advantages:

- **Automation** — Reduces manual infrastructure work.
- **Consistency** — Helps maintain the same infrastructure configuration across environments.
- **Repeatability** — The same infrastructure can be recreated from the configuration.
- **Version Control** — Terraform configuration can be stored in Git.
- **Change Management** — Infrastructure changes can be controlled through code.
- **Collaboration** — Teams can work together on infrastructure configuration.
- **Reusability** — Terraform configurations and modules can be reused.
- **Scalability** — Terraform can manage small as well as large infrastructure environments.
- **Multi-platform support** — Terraform can work with AWS, Azure, Google Cloud, Kubernetes, and many other providers.

When should we use Terraform?
Terraform is useful when infrastructure needs to be:

- Created repeatedly
- Managed across multiple environments
- Managed by multiple team members
- Version-controlled
- Automated through CI/CD
- Consistently configured
- Changed frequently

For very small and temporary infrastructure, manually creating resources may sometimes be simpler. Terraform becomes more valuable as the infrastructure and operational requirements grow.
What problem does Terraform solve?
The main problem Terraform solves is **manual and inconsistent infrastructure management**.
Without Terraform, engineers may manually create resources and configure them differently across environments.
With Terraform, the infrastructure configuration is written as code, so the desired infrastructure can be defined and managed in a consistent way.

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4. What is Terraform Architecture?
Definition: Terraform architecture describes how Terraform processes configuration and communicates with external platforms through providers to create and manage infrastructure.
Simple meaning:
Terraform architecture explains how Terraform connects our Terraform configuration with the actual infrastructure.
Terraform itself does not directly understand how to create every type of cloud resource.
Instead, Terraform uses **providers** to communicate with platforms such as AWS, Azure, Google Cloud, Kubernetes, and other services.

The provider communicates with the platform's APIs and performs the required operations.

Main components of Terraform architecture

The important components are:

- **Terraform Configuration** — Defines the infrastructure we want.
- **Terraform Core** — Processes the configuration, evaluates dependencies, creates the execution plan, and manages Terraform's overall workflow.
- **Provider** — Allows Terraform to communicate with an external platform or service.
- **State** — Stores information about infrastructure managed by Terraform.
- **Infrastructure** — The actual resources created and managed on the target platform.

How does Terraform architecture work?

The process generally works like this:

Terraform Configuration
↓
Terraform Core
↓
Provider
↓
Provider API
↓
Cloud / External Platform
↓
Infrastructure

For example, when Terraform needs to manage an AWS resource, Terraform uses the AWS provider. The provider communicates with AWS APIs and performs the required operation.

Why is the provider important?

The provider acts as the connection between Terraform and the platform that Terraform needs to manage.
Different platforms require different providers.
For example:

- AWS → AWS Provider
- Azure → Azure Provider
- Google Cloud → Google Provider
- Kubernetes → Kubernetes Provider

The provider contains the functionality required to communicate with that platform and manage its supported resources.

What is Terraform Core responsible for?

Terraform Core is responsible for processing the Terraform configuration and determining the required infrastructure changes.
It works with configuration, state, dependencies, and providers to determine what actions Terraform needs to perform.

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5. What is Terraform Workflow?

Definition: Terraform Workflow is the sequence of steps followed to initialize, validate, plan, apply, and manage infrastructure using Terraform.

Simple meaning:

Terraform workflow is the normal process we follow when working with Terraform.
We first prepare the Terraform project, check the configuration, review the changes Terraform wants to make, and then apply those changes to the infrastructure.
The commonly used workflow is:

Write → Initialize → Validate → Plan → Apply → Manage

Step 1: Write Terraform Configuration

First, we define the required infrastructure in Terraform configuration files.
These files describe what infrastructure we want Terraform to create and manage.

Step 2: Initialize Terraform

Terraform needs to prepare the working directory and download the required providers and modules.
This is done using the initialization step.
The main purpose is to make the Terraform project ready for further operations.

Step 3: Validate the Configuration

After initialization, we can check whether the Terraform configuration is syntactically valid and internally consistent.
This helps identify configuration problems before making infrastructure changes.

Step 4: Create a Plan

Terraform then evaluates the configuration and determines what changes are required.
The plan allows us to review whether Terraform intends to:

- Create resources
- Update resources
- Replace resources
- Destroy resources

The important point is that planning is used to **review changes before applying them**.

Step 5: Apply the Changes

After reviewing the plan, we apply the configuration.
Terraform communicates with the required provider and makes the planned infrastructure changes.

Step 6: Manage the Infrastructure

After infrastructure is created, Terraform can continue to manage it.
If the configuration changes later, Terraform can determine what changes are required and update the infrastructure accordingly.
Why is Terraform Workflow important?
Terraform workflow provides a controlled way to manage infrastructure.
Instead of directly making changes without reviewing them, we can first understand what Terraform intends to change and then apply those changes.
This helps reduce unexpected infrastructure changes and makes infrastructure management more predictable.
Important interview flow:
Terraform Code
↓
Initialize
↓
Validate
↓
Plan
↓
Review Changes
↓
Apply
↓
Infrastructure

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6. What is a Terraform Configuration File?

Definition: A Terraform Configuration File is a file used to define infrastructure and Terraform settings using Terraform configuration language. Terraform configuration files normally use the .tf file extension.
Simple meaning:
Terraform configuration file means the file where we write what infrastructure we want Terraform to create and manage.
Terraform configuration can contain:

- Providers — Defines which platform Terraform should communicate with.
- Resources — Defines infrastructure that Terraform should create or manage.
- Variables — Defines values that can be changed or reused.
- Outputs — Displays useful information after infrastructure is created.
- Locals — Defines reusable values inside the configuration.
- Modules — Allows reusable Terraform configurations.

Terraform reads the .tf files in the working directory and uses them as the desired infrastructure configuration.
Why do we use Terraform Configuration Files?
We use configuration files because infrastructure can be defined as code instead of being created manually.
This gives us:

- Reusable infrastructure configuration.
- Version control through Git.
- Easier infrastructure changes.
- Consistent environments.
- Easier collaboration between team members.
- Ability to review infrastructure changes before applying them.

Where are they used?
Terraform configuration files are used in every Terraform project where we define and manage infrastructure.
Common files include:

- main.tf — Main infrastructure configuration.
- variables.tf — Input variable definitions.
- outputs.tf — Output definitions.
- providers.tf — Provider configuration.

Main point:
Terraform Configuration File = .tf file used to define and manage infrastructure and Terraform configuration.

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7. What is a Terraform Provider?
Definition: A Terraform Provider is a component that allows Terraform to communicate with an external platform or service through its APIs and manage resources on that platform.
Simple meaning:
Provider acts as a connection between Terraform and the platform where we want to create or manage infrastructure.
Terraform itself does not directly know how to communicate with AWS, Azure, Google Cloud, Kubernetes, or other platforms.
The provider contains the required logic to communicate with those platforms.
Why do we use Providers?
We use providers because Terraform needs a way to communicate with external platforms and their APIs.
A provider allows Terraform to:

- Authenticate with a platform.

- Communicate with its APIs.
- Create resources.
- Read existing resources.
- Update resources.
- Delete resources.

Common Terraform Providers:

- AWS Provider — For AWS infrastructure.
- AzureRM Provider — For Microsoft Azure.
- Google Provider — For Google Cloud.
- Kubernetes Provider — For Kubernetes resources.
- GitHub Provider — For GitHub resources.

Where are Providers used?
Providers are used whenever Terraform needs to manage resources belonging to an external platform or service.
Main point:
Terraform Provider = Connects Terraform with an external platform and allows Terraform to manage its resources.

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8. What is a Terraform Resource?
Definition: A Terraform Resource represents an infrastructure object that Terraform creates and manages.
Simple meaning:
Resource means the actual infrastructure component that we want Terraform to manage.
A resource can represent:

- EC2 instance.
- VPC.
- Subnet.
- Security Group.
- Load Balancer.
- Database.
- S3 bucket.
- Kubernetes deployment.
- DNS record.

Resources are the main building blocks of Terraform infrastructure configuration.
Why do we use Resources?
We use resources to tell Terraform exactly which infrastructure objects need to be created, updated, or managed.
Terraform uses the resource configuration to understand:

- What type of infrastructure is required.
- Which provider should manage it.
- What configuration the resource should have.
- Which other resources it depends on.

How does a Resource work?
Terraform reads the resource configuration and communicates with the appropriate provider.
The provider then communicates with the actual platform API to create or manage that resource.
Where are Resources used?
Resources are used whenever we want Terraform to create or manage an infrastructure object.
Main point:
Terraform Resource = Represents an actual infrastructure object managed by Terraform.

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9. What is a Terraform Variable?
Definition: A Terraform Variable is an input value that allows us to pass configurable and reusable values into a Terraform configuration instead of hard-coding them.
Simple meaning:
Variable means keeping values separately so the same Terraform code can be reused with different values.
Without variables, we may have to directly write values inside the Terraform configuration.
With variables, those values can be changed without changing the main infrastructure logic.
Why do we use Variables?
Variables are mainly used to make Terraform configurations:

- Reusable.
- Flexible.
- Easier to maintain.
- Easier to manage across environments.
- Less dependent on hard-coded values.

Common values managed using variables include:

- Region.
- Instance type.

- Environment name.
- Server count.
- Resource names.
- CIDR blocks.
- Tags.

Where are Variables used?
Variables are commonly used when the same Terraform configuration needs to work with different values.
For example, Development and Production may use the same infrastructure code but different instance types or server counts.
Main point:
Terraform Variable = An input value used to make Terraform configuration reusable and configurable.

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10. What is a Terraform Output?
Definition: A Terraform Output is used to expose or display useful information produced by Terraform-managed infrastructure.
Simple meaning:
Output means showing useful information about resources after Terraform creates or manages them.
Outputs can provide values such as:

- Public IP address.
- Private IP address.
- DNS name.
- Resource ID.
- Load balancer address.
- Network ID.

Why do we use Outputs?
We use outputs when information generated by one part of the Terraform configuration needs to be easily accessed or exposed.
Outputs are useful for:

- Displaying important resource information.
- Passing information from a child module to its calling module.
- Making resource information available to other Terraform configurations through appropriate mechanisms.
- Avoiding the need to manually search for resource details.

Where are Outputs used?
Outputs are commonly defined in outputs.tf and are especially useful in modular Terraform projects.
Main point:
Terraform Output = Exposes useful information generated by Terraform-managed infrastructure.

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11. What are Terraform Local Values?
Definition: Terraform Local Values are named values defined inside a Terraform configuration that can be reused multiple times within that configuration.
Simple meaning:
Local value means creating a reusable name for a value or expression so we do not have to repeat the same value or calculation throughout the configuration.
A local value is only available inside the Terraform module where it is defined.
Why do we use Local Values?
We use local values to:

- Avoid repeating the same expression.
- Make configurations easier to read.
- Simplify complex expressions.
- Create common naming or tagging logic.
- Reuse calculated values within a configuration.

Important difference:
Variables are generally used to accept input from outside the module.
Locals are used to create reusable values inside the configuration itself.
Where are Local Values used?
They are useful when multiple resources need to use the same calculated value, naming pattern, tags, or configuration logic.
Main point:
Local Value = A reusable named value defined inside a Terraform configuration.

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12. What is a Terraform Data Source?
Definition: A Terraform Data Source is used to read information about existing resources or external data without creating that resource through the data source.
Simple meaning:
Data source means Terraform can read information that already exists and use that information in its configuration.
The important point is that the data source **reads** information; it does not create the resource it is reading.
Why do we use Data Sources?
We use data sources when required infrastructure already exists and Terraform needs information about it.
They are useful when:

- A resource was created outside the current Terraform configuration.
- Another Terraform configuration manages the resource.
- We need to retrieve existing infrastructure information.
- We need IDs or other attributes of existing resources.

What can Terraform read?
Terraform can retrieve information such as:

- Existing VPC.
- Existing subnet.
- Existing AMI.
- Existing DNS information.
- Existing cloud resources supported by the provider.

Where are Data Sources used?
Data sources are commonly used when Terraform-managed resources need to reference existing infrastructure.
Main point:
Data Source = Reads information about existing resources so Terraform can use that information.

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13. What is Terraform State?
Definition: Terraform State is the information Terraform maintains about the infrastructure resources it manages and their relationship with the Terraform configuration.
Simple meaning:
Terraform state is Terraform's record of the resources it is managing.
Terraform uses state to understand:

- Which resources it manages.
- The resource addresses in the configuration.
- Important resource attributes.
- The relationship between configuration and real infrastructure.

State helps Terraform determine what has already been created and what changes may be required.
Why do we use Terraform State?
Terraform needs state to track managed infrastructure and calculate changes efficiently.
When the configuration changes, Terraform uses state together with information from the provider to determine what actions are required.
State is therefore an important part of Terraform's infrastructure management process.
Where is State stored?
State can be stored:

- Locally.
- In a remote backend.

For team environments, remote state is commonly preferred because multiple users and automation systems may need access to the same state.
Important:
Terraform state can contain sensitive information depending on the resources being managed, so it must be properly protected.
Main point:
Terraform State = Keeps track of Terraform-managed infrastructure and helps Terraform determine required changes.

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14. What is a Terraform State File?
Definition: A Terraform State File is the file used to store Terraform state information when Terraform is using local state storage. The default local state file is terraform.tfstate.
Simple meaning:
Terraform state file is the physical file where Terraform stores the state information for the infrastructure it manages when using the local backend.
It contains information Terraform needs to understand the resources recorded in its state.
Why do we use the State File?
Terraform uses the state file to:

- Track managed resources.
- Store resource information.
- Associate Terraform configuration with real infrastructure.
- Determine changes during planning.
- Maintain the current Terraform state.

Important:
The state file should not be treated like normal application code.
It may contain sensitive information, depending on the configuration and provider, so it should be protected from unauthorized access.
Where is it stored?
With local state, it is normally stored in the Terraform working directory as:
terraform.tfstate
With remote state, the state is stored according to the configured backend.
Main point:

Terraform State File = Stores Terraform state information used to track and manage infrastructure.

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15. What is a Terraform Backend?
Definition: A Terraform Backend defines where Terraform stores its state and how Terraform accesses that state.
Simple meaning:
Backend means the place and mechanism Terraform uses to store and access its state.
Terraform can use a local backend or a remote backend.
Why do we use a Backend?
A backend is important because Terraform needs a reliable location to store state.
In team environments, a remote backend can provide shared state access and, depending on the backend, additional capabilities such as state locking and access control.
This helps multiple developers and CI/CD systems work with the same infrastructure state.
Types of Backend:

- Local Backend — Stores state locally.
- Remote Backend — Stores state in a remote location.

Where is a Backend used?
A local backend may be suitable for learning or small personal projects.
Remote backends are commonly used in team and production environments where shared state and controlled access are required.
Main point:
Terraform Backend = Defines where Terraform state is stored and how Terraform accesses it.

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16. What is a Local Backend?
Definition: A Local Backend is the default Terraform backend that stores Terraform state on the local machine where Terraform is executed.
Simple meaning:
Local backend means Terraform stores its state locally instead of storing it in a shared remote location.
With the local backend, Terraform normally stores state in the Terraform working directory.
Why do we use a Local Backend?
A local backend is simple to configure and useful for:

- Learning Terraform.
- Testing.
- Small personal projects.
- Temporary development environments.

It does not require setting up a separate remote state storage system.
Limitation:
Local state is not ideal for team environments because the state is tied to the local machine unless it is otherwise shared.
This can make collaboration and centralized state management difficult.
Where is it used?
It is mainly suitable for individual development, learning, testing, and small projects.
Main point:
Local Backend = Stores Terraform state locally on the machine running Terraform.

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17. What is a Remote Backend?
Definition: A Remote Backend is a Terraform backend that stores Terraform state in a remote location instead of storing it only on the local machine.
Simple meaning:
Remote backend means Terraform state is stored in a centralized remote location so authorized team members and automation systems can access the same state.
Why do we use a Remote Backend?
Remote backends are useful for team-based infrastructure management because they can provide:

- Shared state.
- Centralized state storage.
- Controlled access.
- State locking, when supported by the backend.
- Better integration with CI/CD systems.
- Better protection of state when properly configured.

When do we use it?
Remote backends are especially useful when:

- Multiple developers manage the same infrastructure.
- CI/CD pipelines manage infrastructure.
- Production infrastructure is managed by a team.
- Centralized state management is required.

Where is it used?
It is commonly used in company and production Terraform environments.
Main point:
Remote Backend = Stores Terraform state remotely for shared and controlled infrastructure management.

18. What is terraform init**?**
Definition: terraform init is used to initialize a Terraform working directory and prepare it for Terraform operations.
Simple meaning:
terraform init is normally the first Terraform command we run in a new or changed Terraform project.
It prepares Terraform by installing or initializing the dependencies required by the configuration.
What does terraform init **do?**
It can:

- Initialize the Terraform working directory.
- Download required providers.
- Initialize configured modules.
- Configure or initialize the backend.
- Use the provider dependency information recorded in the lock file.

Why do we use terraform init**?**
Terraform needs the required providers, modules, and backend configuration before it can properly perform operations such as planning and applying infrastructure.
When do we use it?
We commonly run it:

- When starting a new Terraform project.
- After cloning a Terraform project.
- After adding or changing provider requirements.
- After adding or changing modules.
- After changing backend configuration.

Main point:
terraform init = Initializes the Terraform project and installs/configures its required dependencies.

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19. What is terraform plan**?**
Definition: terraform plan is used to preview the infrastructure changes Terraform intends to make without normally applying those changes.
Simple meaning:
terraform plan shows us what Terraform is planning to do before we actually change the infrastructure.
Terraform evaluates the configuration, state, and information available from the provider to determine the required actions.
The plan can show resources that Terraform intends to:

- Create.
- Update.
- Replace.
- Destroy.

Why do we use terraform plan**?**
We use it to review infrastructure changes before applying them.
It helps us:

- Detect unexpected changes.
- Review planned infrastructure modifications.
- Catch configuration mistakes.
- Safely review changes before execution.
- Use plans as part of CI/CD and approval workflows.

When do we use it?
We normally use terraform plan after making Terraform configuration changes and before terraform apply.
Important:
terraform plan normally does not make the infrastructure changes itself. It only shows the proposed changes.
Main point:
terraform plan = Shows the changes Terraform intends to make before applying them.

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20. What is terraform apply**?**
Definition: terraform apply is used to apply the Terraform configuration and make the required infrastructure changes.
Simple meaning:
terraform apply is the command that actually performs the planned infrastructure changes.
Terraform determines the required actions and communicates with the appropriate provider to create, update, replace, or delete resources.
Why do we use terraform apply**?**
We use terraform apply when we are ready to make the Terraform configuration changes to the actual infrastructure.
It is used to:

- Create infrastructure.
- Update infrastructure.
- Replace resources when required.
- Delete resources when the configuration requires it.

When do we use it?

We normally run terraform apply after reviewing the planned changes and confirming that the changes are correct.
In production environments, it is commonly executed through an approved and controlled workflow.
Important:
Unlike terraform plan, terraform apply actually changes the infrastructure.
Main point:

1. terraform apply = Applies the Terraform configuration and makes the required infrastructure changes.

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21. What is terraform destroy**?**

Definition: terraform destroy is a Terraform command used to destroy the infrastructure managed by the current Terraform configuration.
Simple meaning:
It tells Terraform to remove the resources that it manages.
Terraform identifies the resources that need to be destroyed and performs the required deletion through the configured provider.
Why do we use terraform destroy**?**
It is mainly used when infrastructure is no longer required, such as temporary environments or resources created for testing.
How does it work?
Terraform uses the configuration and state information to determine which managed resources need to be removed.
Terraform
↓
terraform destroy
↓
Terraform Plan
↓
Provider
↓
Resources Destroyed
Important points:

- It can remove multiple Terraform-managed resources.
- It should be used carefully.
- Destroying a resource can result in permanent data loss.
- Production resources should not be destroyed without proper verification and approval.

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22. What is terraform validate**?**
Definition: terraform validate is a Terraform command used to check whether the Terraform configuration is syntactically valid and internally consistent.
Simple meaning:
It checks whether the Terraform configuration is written correctly before we proceed with infrastructure changes.
Why do we use terraform validate**?**
It helps identify configuration problems early.
It can detect issues related to:

- Invalid Terraform syntax
- Incorrect configuration structure
- Invalid arguments
- Incorrect references within the configuration

When do we use it?
It is commonly used after modifying Terraform code and before creating or applying infrastructure changes.
Important points:

- It validates the Terraform configuration.
- It does not create infrastructure.
- It does not modify infrastructure.
- It is commonly used in CI/CD pipelines.

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23. What is terraform fmt**?**
Definition: terraform fmt is a Terraform command used to automatically format Terraform configuration files according to Terraform's standard formatting style.
Simple meaning:
It makes Terraform code clean, consistent, and properly formatted.
Why do we use terraform fmt**?**
Different developers may write Terraform code with different spacing and formatting.
terraform fmt standardizes the formatting so the code remains consistent across the project.
When do we use it?
It is commonly used after writing or modifying Terraform configuration and before committing code to Git.
Important points:

- It formats Terraform files automatically.
- It improves code readability.
- It keeps formatting consistent across a team.
- It does not validate whether the infrastructure configuration is logically correct.
- It does not create or modify infrastructure.

24. What is terraform show**?**
Definition: terraform show is a Terraform command used to display the details of the current Terraform state or a saved Terraform plan in a human-readable format.
Simple meaning:
It allows us to see what information Terraform currently has about the managed infrastructure.
Why do we use terraform show**?**
It is useful for:

- Inspecting Terraform state.
- Reviewing resource information.
- Understanding what Terraform knows about managed resources.
- Inspecting a saved plan.

When do we use it?
It is mainly useful during troubleshooting, state inspection, and infrastructure review.
Important points:

- It can display current state information.
- It can display information from a saved plan.
- It is mainly an inspection command.
- It does not create or destroy infrastructure.

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25. What are Terraform State Commands?
Definition: Terraform State Commands are commands used to inspect and manage resources recorded in Terraform state.
Simple meaning:
These commands are used when we need to work directly with the information Terraform maintains in its state.
Why do we use State Commands?
They are useful when we need to:

- List resources in state.
- Inspect a resource.
- Move a resource to another state address.
- Remove a resource from state.

Important State Commands:
terraform state list
Lists resources currently recorded in Terraform state.
terraform state show
Displays detailed state information for a specific resource.
terraform state mv
Moves a resource from one state address to another.
terraform state rm
Removes a resource from Terraform state.
Important point:
State commands should be used carefully because they directly modify or inspect Terraform's state information.

---

26. What are Terraform Dependencies?
Definition: Terraform Dependencies define relationships between resources when one resource depends on another resource.
Simple meaning:
A dependency tells Terraform that one resource needs another resource to be created or handled first.
Terraform uses dependencies to determine the correct order of resource operations.
Why are Dependencies important?
They help Terraform understand:

- Which resource should be created first.
- Which resource depends on another.
- The appropriate order for resource operations.
- The appropriate reverse order during destruction.

How does Terraform identify dependencies?
Terraform can identify dependencies automatically through resource references.
Dependencies can also be explicitly defined when Terraform cannot determine the relationship automatically.
Important points:
There are two main types:

- Implicit Dependency
- Explicit Dependency

---

27. What is an Implicit Dependency?
Definition: An Implicit Dependency occurs when Terraform automatically detects that one resource depends on another through a reference between them.
Simple meaning:
Terraform understands the dependency automatically because one resource uses information from another resource.
For example, when one resource references an attribute of another resource, Terraform can determine that relationship.
Why do we use Implicit Dependencies?
We do not normally need to manually define these dependencies because Terraform can determine them from the configuration.
How does it work?

Terraform creates a dependency relationship from the reference.
Resource A
↓
Referenced Attribute
↓
Resource B
Terraform then uses that relationship to determine the appropriate operation order.
Important point:
Implicit dependencies are generally preferred because Terraform can understand them directly from the configuration.

---

28. What is an Explicit Dependency?
Definition: An Explicit Dependency is a dependency that is manually specified using the depends_on meta-argument.
Simple meaning:
We explicitly tell Terraform that one resource depends on another when Terraform cannot automatically identify the dependency.
Why do we use Explicit Dependencies?
They are useful when a dependency exists logically but there is no direct resource attribute reference that Terraform can use to detect it.
How do we define it?
Terraform provides the depends_on meta-argument for explicitly declaring dependencies.
Resource A
↓
depends_on
↓
Resource B
Terraform then knows that the required dependency must be handled first.
Important points:

- depends_on creates an explicit dependency.
- It should be used only when necessary.
- Unnecessary dependencies can make Terraform operations more restrictive.

---

29. What are Terraform Expressions?
Definition: Terraform Expressions are used to produce values within Terraform configuration.
Simple meaning:
Expressions allow Terraform configuration to calculate or reference values dynamically instead of using only fixed values.
Expressions can use:

- Variables
- Resources
- Local values
- Functions
- Operators
- Conditional expressions
- Other Terraform values

Why do we use Expressions?
They make Terraform configuration more dynamic and reusable.
Instead of hard-coding every value, expressions allow Terraform to derive values from existing configuration information.
How do Expressions work?
Variables / Resources / Functions
↓
Expression
↓
Value
Important point:
Expressions are a fundamental part of Terraform's configuration language and are used throughout Terraform configurations.

---

30. What are Terraform Conditional Expressions?
Definition: Terraform Conditional Expressions allow Terraform to select one value based on whether a condition is true or false.
Simple meaning:
A conditional expression allows Terraform to make a decision between two possible values.
The basic concept is:
Condition
↓
┌──┴──┐
True False
↓ ↓
Value A Value B
Why do we use Conditional Expressions?
They are useful when configuration needs to behave differently based on:

- Environment
- Variables
- Configuration values

- Other conditions

When do we use them?
They are commonly used when different environments or configurations require different values.
Important point:
Conditional expressions help reduce duplicate configuration by allowing values to be selected dynamically.

---

31. What are Terraform Functions?
Definition: Terraform Functions are built-in operations that transform, calculate, or process values within Terraform configuration.
Simple meaning:
Functions help Terraform perform operations on values instead of manually calculating or modifying them.
Terraform provides functions for areas such as:

- String manipulation
- Number calculations
- Collection operations
- Type conversion
- Searching and selecting values
- Data transformation

Why do we use Functions?
Functions help make Terraform configurations:

- Dynamic
- Reusable
- Flexible
- Easier to maintain

How do Functions work?
Input Value
↓
Terraform Function
↓
Processed Value
Important point:
Terraform provides built-in functions; we do not define custom Terraform functions in the same way we define functions in general-purpose programming languages.

---

32. What is a Terraform List?
Definition: A Terraform List is an ordered collection of values where each element has a position or index.
Simple meaning:
A list stores multiple values in a specific order.
The position of each element matters, and elements can be accessed using their index.
Why do we use Lists?
Lists are useful when:

- Order matters.
- We need multiple values.
- We need to access values by position.

Important characteristics:

- Ordered collection.
- Elements have indexes.
- Duplicate values can exist.
- Elements are accessed based on position.

Simple structure:
List
↓
Index 0
Index 1
Index 2

---

33. What is a Terraform Map?
Definition: A Terraform Map is a collection of key-value pairs where each key is associated with a value.
Simple meaning:
A map stores values using meaningful keys instead of numeric positions.
The key is used to identify and access the corresponding value.
Why do we use Maps?
Maps are useful when we want to associate:

- Names with values.
- Environments with configurations.
- Resource identifiers with properties.
- Keys with specific settings.

Important characteristics:

- Contains key-value pairs.
- Values are accessed using keys.

- Keys provide meaningful identification.
- Useful for configuration-based lookups.

Simple structure:

Key → Value
Key 1 → Value 1
Key 2 → Value 2
Key 3 → Value 3

---

34. What is a Terraform Set?

Definition: A Terraform Set is an unordered collection of unique values.

Simple meaning:

A set stores values where duplicates are not retained and element position is not used as the identity.

Why do we use Sets?

Sets are useful when:

- We need unique values.
- Order is not important.
- Duplicate values should not be retained.

Important characteristics:

- Unordered collection.
- Contains unique values.
- Duplicate values are removed.
- Elements should not be treated as having stable numeric positions like list elements.

List vs Set:

List
→ Ordered
→ Index-based

Set
→ Unordered
→ Unique values

---

35. What is a Terraform Object?

Definition: A Terraform Object is a structured value containing multiple named attributes, where each attribute can have its own specific type.

Simple meaning:

An object allows related information to be grouped together under named attributes.
Instead of treating every value separately, an object represents a structured piece of information.

Why do we use Objects?

Objects are useful when:

- Multiple related values need to be grouped.
- Different attributes have different types.
- We want structured input variables.
- We want clearly defined data structures.

Important characteristics:

- Contains named attributes.
- Each attribute can have its own type.
- Provides a structured way to represent related information.

Simple structure:

Object
├— Attribute 1
├— Attribute 2
└— Attribute 3

---

36. What is a Terraform Tuple?

Definition: A Terraform Tuple is an ordered collection of elements where each position can have a different type.

Simple meaning:

A tuple is similar to a list because order matters, but the elements at different positions can have different types.

Why do we use Tuples?

Tuples are useful when we need:

- Ordered values.
- Multiple values of different types.
- A fixed structure where each position has a defined type.

Important characteristics:

- Ordered collection.
- Each position can have a different type.
- The number and type of elements can be defined as part of the tuple type.

Tuple vs List:

List
→ Ordered

→ Elements generally use the same type

Tuple
→ Ordered
→ Elements can use different types

---

37. What is a Terraform for **Expression?**
Definition: A Terraform for expression is used to iterate over elements of a collection and produce a new value.
Simple meaning:
A for expression allows us to process multiple values and create another collection from them.
It can be used to create:

- Lists
- Sets
- Maps

from existing collections.
Why do we use for **Expressions?**
They are useful when we need to:

- Transform values.
- Filter values.
- Create new collections.
- Process multiple elements dynamically.

How does it work?
Existing Collection
↓
for Expression
↓
Transformed Collection
Important point:
A for expression is mainly used for transforming or filtering collection values. It is different from resource-level iteration using count or for_each.

---

38. What is Terraform count**?**
Definition: count is a Terraform meta-argument used to create multiple instances of a resource or module based on a numeric value.
Simple meaning:
count tells Terraform how many instances of a resource or module should exist.
Why do we use count**?**
It is useful when the required number of similar instances is primarily determined by a number.
How does it work?
Terraform creates instances using numeric indexes.
count = N
↓
Resource 0
Resource 1
Resource 2
...
Each instance can be referenced using its index.
Important points:

- count uses a number.
- Instances are identified using indexes.
- It is useful for quantity-based resources.
- Changing the index structure can sometimes cause resource replacement or address changes.

---

39. What is Terraform for_each**?**
Definition: for_each is a Terraform meta-argument used to create multiple instances of a resource or module based on the elements of a map or set.
Simple meaning:
for_each creates one instance for each element in a map or set.
Unlike count, instances are identified using meaningful keys.
Why do we use for_each**?**
It is useful when:

- Each instance needs a meaningful identity.
- Instances have different configurations.
- We want to manage individual instances using keys.
- We are working with maps or sets.

How does it work?
Map / Set
↓
for_each
↓

Multiple Instances
↓
Key-based Addresses

Important points:

- for_each works with maps and sets.
- Instances are identified using keys.
- It is generally better than count when individual instances have meaningful identities.
- It helps avoid relying only on numeric indexes.

---

40. What is the difference between count **and** for_each**?**

Definition: count and for_each are Terraform meta-arguments used to create multiple instances of resources or modules, but they identify and manage those instances differently.

Simple meaning:

count = Create resources based on a number.
for_each = Create resources based on a map or set.

count

- Uses a number.
- Creates instances using numeric indexes.
- Best when resources are mainly quantity-based.
- Instances are identified as 0, 1, 2, and so on.

for_each

- Uses a map or set.
- Creates one instance for each element.
- Uses keys to identify instances.
- Better when each instance has a meaningful identity.

Main difference:

count
↓
Number
↓
0, 1, 2, 3

for_each
↓
Map / Set
↓
Meaningful Keys

When should we use which?

Use count when the main requirement is **how many instances** are needed.
Use for_each when the main requirement is **which individually identifiable instances** are needed.

Important interview point:

1. for_each is often preferred when resources have meaningful identities because changes to one key do not depend on shifting numeric indexes in the same way that count instances can.

---

42. What are Terraform Dynamic Blocks?

Definition:

Terraform Dynamic Blocks are used to dynamically generate repeated nested configuration blocks inside a resource based on a collection such as a list, set, or map.

Simple meaning:

A dynamic block means creating multiple similar nested blocks automatically instead of writing each block manually.

Why do we use Dynamic Blocks?

Dynamic blocks are useful when:

- A resource contains repeated nested blocks.
- The number of nested blocks can change.
- The values come from variables, lists, maps, or other collections.
- We want to avoid repeating the same configuration.

How do Dynamic Blocks work?

Terraform takes a collection and generates a nested block for each element in that collection.

Simple flow:

Collection
↓
Dynamic Block
↓
Multiple Nested Blocks
↓
Resource

Important interview point:

Dynamic blocks are mainly used for **repeated nested blocks**, not for creating multiple resources.

43. What is a Terraform Module?

Definition:

A Terraform Module is a collection of Terraform configuration files that are grouped together to create reusable and organized infrastructure code.

Simple meaning:

A module is a **reusable Terraform code package**.

Instead of writing the same infrastructure configuration again and again, we can create it once inside a module and reuse it.

Why do we use Modules?

Modules help us:

- Reuse Terraform code.
- Avoid duplicate configuration.
- Organize large Terraform projects.
- Standardize infrastructure.
- Separate infrastructure into logical components.
- Maintain infrastructure more easily.

How do Modules work?

A module can contain:

- Resources
- Variables
- Outputs
- Local values
- Other Terraform configuration

A root configuration can call the module and provide required input values.

Simple flow:

Root Module
↓
Child Module
↓
Resources
↓
Infrastructure

Important interview point:

Module = **Reusable Terraform configuration.**

---

44. What is a Root Module?

Definition:

The Root Module is the main Terraform module represented by the Terraform configuration files in the directory from which Terraform commands are executed.

Simple meaning:

The root module is the **main Terraform project configuration**.

Every Terraform configuration has a root module.

What can a Root Module contain?

It can contain:

- Resources
- Variables
- Outputs
- Providers
- Locals
- Module calls
- Other Terraform configuration

The root module can also call child modules.

How does it work?

When we run Terraform commands from a directory, the .tf files in that directory form the root module.

Simple flow:

Terraform Project Directory
↓
Root Module
↓
Resources / Child Modules
↓
Infrastructure

Important interview point:

Root Module = **The main module from which Terraform execution starts.**

---

45. What is a Child Module?

Definition:

A Child Module is a Terraform module that is called by another module, such as the root module or another child module.

Simple meaning:

A child module is a **reusable module used by another Terraform module**.

Why do we use Child Modules?

They help us:

- Break large Terraform configurations into smaller parts.
- Reuse infrastructure code.
- Organize infrastructure logically.
- Maintain large projects more easily.
- Separate components such as networking, compute, and databases.

How does it work?

The root module calls a child module and passes required input values to it.
The child module creates its resources and can return values through outputs.

Simple flow:

Root Module
↓
Child Module
↓
Resources

Important interview point:

Child Module = **A module called by another module.**

---

46. What are Module Inputs and Outputs?

Definition:

Module Inputs and Outputs are used to pass information into and out of Terraform modules.

Simple meaning:

- **Input = Information given to a module.**
- **Output = Information returned from a module.**

Module Inputs

Module inputs are usually defined using input variables.
They allow the calling module to provide values such as:

- Region
- Environment
- Instance type
- Resource names
- CIDR ranges

Module Outputs

Outputs expose useful information created or calculated by the module.
Examples include:

- Resource IDs
- IP addresses
- DNS names
- Network IDs

How does it work?

Calling Module
↓
Input Variables
↓
Child Module
↓
Resources
↓
Outputs
↓
Calling Module

Important interview point:

Inputs provide values **to** a module, while outputs expose values **from** a module.

---

47. What is Terraform Registry?

Definition:

Terraform Registry is a platform where Terraform providers and reusable modules can be discovered and used.

Simple meaning:

Terraform Registry is like a **repository for Terraform providers and modules**.

Why do we use Terraform Registry?

It helps us:

- Find existing modules.
- Find providers.
- Reuse community or official modules.
- Avoid writing common infrastructure from scratch.
- Discover available Terraform components.

What can we find in the Registry?

- Providers
- Modules
- Provider documentation
- Module documentation
- Usage information

How do we use it?
We select a required provider or module and reference it from our Terraform configuration.
Simple flow:
Terraform Registry
↓
Provider / Module
↓
Terraform Configuration
↓
Infrastructure
Important interview point:
Terraform Registry = **A platform for discovering and using Terraform providers and modules.**

---

48. What are Terraform Workspaces?
Definition:
Terraform Workspaces allow us to maintain multiple separate state instances for the same Terraform configuration.
Simple meaning:
A workspace allows the **same Terraform code to use different Terraform state**.
Each workspace has its own state.
Why do we use Workspaces?
They can be useful when:

- The same configuration needs separate states.
- We need isolated infrastructure instances.
- We want simple separation between environments or deployments.

How do Workspaces work?
The Terraform configuration remains the same, but each workspace maintains separate state.
For example:

- Development workspace
- Testing workspace
- Production workspace

Each workspace has its own state.
Simple flow:
Same Terraform Configuration
↓
Workspace
↓
Separate State
Important interview point:
Workspace = **Separate state for the same Terraform configuration.**
Interview note:
Workspaces are not always the best choice for separating production environments in large organizations. Teams often use separate configurations, modules, or backend/state structures for stronger isolation.

---

49. What are Terraform Provisioners?
Definition:
Terraform Provisioners are used to execute commands or scripts on a local machine or remote resource during resource creation or destruction.
Simple meaning:
Provisioners allow Terraform to **run commands or scripts as part of a resource lifecycle**.
Why do we use Provisioners?
They can be used when:

- A resource requires a special initialization step.
- A required task cannot be handled better through a provider resource or another configuration mechanism.
- We need to execute a local or remote command.

Types of Provisioners include:

- local-exec
- remote-exec
- file

Important point:
Provisioners should generally be considered a **last resort** because they can be less predictable and harder to manage than native Terraform resources or configuration-management tools.
Simple flow:
Terraform
↓
Create Resource

↓
Provisioner
↓
Execute Command / Script
Important interview point:
Provisioner = **Runs commands or scripts during resource lifecycle operations and should generally be used as a last resort.**

---

50. What is Terraform Import?
Definition:
Terraform Import is used to bring an existing infrastructure resource under Terraform management by associating it with a Terraform resource address and recording it in Terraform state.
Simple meaning:
Import is used when a resource **already exists**, but Terraform is not currently managing it.
Why do we use Import?
Import is useful when:

- Infrastructure was created manually.
- Infrastructure was created using another tool.
- We want to start managing an existing resource with Terraform.
- We do not want to recreate an existing resource.

Important point:
Import does **not** create the resource.
The resource must already exist.
How does Import work?
Existing Infrastructure
↓
Terraform Resource Configuration
↓
Import
↓
Terraform State
↓
Terraform Management
After importing, we should make sure the Terraform configuration correctly represents the existing resource.
Important interview point:
Terraform Import = **Brings existing infrastructure into Terraform state for management.**

---

51. What are Terraform Lifecycle Rules?
Definition:
Terraform Lifecycle Rules are meta-arguments used to customize how Terraform creates, updates, and destroys resources.
Simple meaning:
Lifecycle rules tell Terraform **how a resource should behave during changes**.
Why do we use Lifecycle Rules?
They are useful when we need special control over:

- Resource replacement.
- Resource destruction.
- Attribute changes.
- Resource creation order.

Important lifecycle rules include:

- create_before_destroy
- prevent_destroy
- ignore_changes
- replace_triggered_by

How do we define them?
Lifecycle rules are normally specified inside a resource's lifecycle block.
Simple flow:
Resource
↓
Lifecycle Rules
↓
Create / Update / Destroy Behavior
Important interview point:
Lifecycle Rules = **Control Terraform's behavior when managing resource changes.**

---

52. What is create_before_destroy?
Definition:
create_before_destroy is a Terraform lifecycle rule that tells Terraform to create a replacement resource before destroying the existing resource when replacement is required.
Simple meaning:
Instead of:
Old Resource → Destroy → New Resource

Terraform attempts:
New Resource → Create → Old Resource → Destroy
Why do we use it?
It can help reduce downtime when:

- A resource must be replaced.
- The platform allows both resources to exist temporarily.
- Availability is important.

Important point:
This does not guarantee zero downtime. The surrounding infrastructure and resource constraints must also support the replacement strategy.
Simple flow:
Existing Resource
↓
Replacement Required
↓
Create New Resource
↓
Destroy Old Resource
Important interview point:
create_before_destroy = **Create the replacement before destroying the existing resource.**

---

53. What is prevent_destroy?
Definition:
prevent_destroy is a Terraform lifecycle rule that prevents Terraform from destroying a resource when Terraform would otherwise attempt to destroy it.
Simple meaning:
It acts as a **protection mechanism against accidental resource destruction**.
Why do we use it?
It is useful for important resources such as:

- Production databases
- Critical storage
- Important infrastructure resources

When prevent_destroy = true is configured, Terraform produces an error if the operation requires destruction of that resource.
How does it work?
Resource
↓
prevent_destroy = true
↓
Destroy Attempt
↓
Terraform Error
↓
Resource Protected
Important interview point:
prevent_destroy = **Prevents Terraform from destroying a protected resource.**

---

54. What is ignore_changes?
Definition:
ignore_changes is a Terraform lifecycle rule that tells Terraform to ignore changes to specified resource attributes when determining whether to update the resource.
Simple meaning:
It tells Terraform:
"For these specific attributes, don't try to manage external changes."
Why do we use it?
It is useful when:

- Some attributes are intentionally changed outside Terraform.
- Another system manages a particular attribute.
- We do not want Terraform to repeatedly revert an external change.

How does it work?
Terraform Resource
↓
External Change
↓
Specified Attribute
↓
ignore_changes
↓
Terraform Ignores That Change
Important point:
ignore_changes should be used carefully because Terraform will intentionally stop reacting to changes in the specified attributes.

Important interview point:
ignore_changes = **Tells Terraform to ignore changes to selected resource attributes.**

---

55. What is the Terraform Lock File?
Definition:
The Terraform Lock File is a file named .terraform.lock.hcl that records the selected provider versions and provider package checksums.
Simple meaning:
The lock file helps Terraform use **consistent provider versions and verify provider packages**.
Why do we use it?
It helps with:

- Provider version consistency.
- Reproducible Terraform runs.
- Provider package verification.
- Team consistency.
- CI/CD consistency.

What does it contain?
The lock file mainly records:

- Provider selections
- Provider versions
- Provider checksums

Important point:
The lock file should generally be committed to version control so the team and CI/CD systems use the same provider selections.
Simple flow:
Terraform Configuration
↓
Provider Selection
↓
.terraform.lock.hcl
↓
Consistent Provider Installation
Important interview point:
Terraform Lock File = **Records provider selections and checksums for consistent provider installation.**

---

56. What is State Locking?
Definition:
State Locking is a mechanism that prevents multiple Terraform operations from modifying the same Terraform state at the same time when the backend supports locking.
Simple meaning:
State locking prevents **two Terraform operations from changing shared state simultaneously**.
Why do we use State Locking?
It helps prevent:

- Concurrent state modifications.
- Conflicting Terraform operations.
- Potential state corruption.
- Multiple users applying changes simultaneously.

How does it work?
Terraform Operation
↓
Acquire State Lock
↓
Perform Operation
↓
Update State
↓
Release Lock
If another operation tries to modify the same locked state, it cannot proceed normally until the lock is released, depending on backend and Terraform behavior.
Important interview point:
State Locking = **Prevents concurrent modifications to shared Terraform state when supported by the backend.**

---

57. What are Sensitive Values in Terraform?
Definition:
Sensitive Values are Terraform values marked as sensitive so Terraform avoids displaying them normally in command output.
Simple meaning:
Sensitive values are values that should **not be openly displayed**, such as:

- Passwords
- API tokens
- Access credentials
- Private keys

- Secrets

Why do we mark values as sensitive?
To reduce accidental exposure in:

- Terraform output
- CLI output
- Logs or displayed values

Important point:
Marking a value as sensitive does **not automatically encrypt it everywhere**.
Sensitive information may still be stored in Terraform state depending on how it is used.
Therefore, the state and backend must also be properly protected.
How does it work?
Secret Value
↓
Sensitive Marking
↓
Hidden / Redacted Output
Important interview point:
Sensitive = **Tells Terraform to treat a value carefully and avoid exposing it normally in output.**

---

58. What is Secrets Management in Terraform?
Definition:
Secrets Management in Terraform is the practice of securely storing, retrieving, and protecting sensitive information used by infrastructure.
Simple meaning:
It means handling secrets such as passwords, API keys, tokens, and certificates **securely instead of exposing them in Terraform code**.
Why is Secrets Management important?
Because infrastructure configurations may require sensitive credentials.
Hard-coding secrets in Terraform files can cause them to be exposed through:

- Git repositories
- Code reviews
- Logs
- State files
- CI/CD systems

How should secrets be handled?
Common approaches include:

- Using a dedicated secret-management system.
- Using protected CI/CD secret variables.
- Restricting access to Terraform state.
- Marking appropriate Terraform values as sensitive.
- Avoiding hard-coded credentials in .tf files.

Important point:
Marking a value as sensitive does not mean the secret is automatically secure.
Terraform state may still contain sensitive information.
Important interview point:
Secrets Management = **Securely handling infrastructure credentials and sensitive data.**

---

59. What is Terraform with AWS?
Definition:
Terraform with AWS means using Terraform and the AWS provider to define, create, and manage AWS infrastructure as code.
Simple meaning:
Terraform can communicate with AWS and manage AWS resources through the **AWS provider**.
What AWS resources can Terraform manage?
For example:

- VPC
- Subnets
- EC2
- Security Groups
- Load Balancers
- RDS
- S3
- IAM
- Route tables
- NAT gateways

How does Terraform communicate with AWS?
Terraform
↓
AWS Provider
↓

AWS APIs
↓
AWS Resources
The AWS provider contains the functionality required for Terraform to interact with AWS services.
Why is this useful?
Instead of manually creating AWS infrastructure through the AWS Console, infrastructure can be defined and managed through Terraform configuration.
Important interview point:
Terraform with AWS = **Using Terraform and the AWS provider to manage AWS infrastructure as code.**

---

60. What is Terraform with CI/CD?
Definition:
Terraform with CI/CD means integrating Terraform into a Continuous Integration and Continuous Delivery pipeline to automate infrastructure validation, planning, review, and deployment.
Simple meaning:
Instead of a developer manually running Terraform commands on a laptop, Terraform can be executed through a controlled CI/CD pipeline.
Why do we use Terraform with CI/CD?
It helps us:

- Automate infrastructure deployments.
- Validate Terraform code automatically.
- Generate plans for review.
- Control production changes.
- Maintain an audit trail through version control and pipeline history.
- Reduce manual infrastructure operations.

How does Terraform CI/CD work?
A typical workflow is:
Git Push
↓
CI/CD Pipeline
↓
terraform fmt
↓
terraform validate
↓
terraform plan
↓
Review / Approval
↓
terraform apply
↓
Infrastructure
Important point:
Production infrastructure changes are commonly protected with review and approval processes rather than allowing every code change to automatically apply.
Important interview point:
Terraform with CI/CD = **Automating and controlling Terraform infrastructure changes through a CI/CD pipeline.**